

Multiple Choice (10 marks)

1. In most ecosystems, net primary productivity is important because it represents the
 - (a) biomass of all producers
 - (b) biomass of all consumers
 - (c) energy available to heterotrophs
 - (d) energy consumed by decomposers
 - (e) total solar energy converted to chemical energy by producers
2. What prevents the stomach from being digested by its own secretions?
 - (a) The stomach has a high permeability to hydrogen ions
 - (b) The presence of beneficial bacteria that neutralize stomach acid
 - (c) The stomach lining is replaced every month
 - (d) The stomach lining secretes digestive enzymes only when food is present
 - (e) The production of bile in the stomach neutralizes the digestive acids
3. In modern terminology, diversity is understood to be a result of genetic variation. Sources of variation for evolution include all of the following except
 - (a) mistakes in translation of structural genes.
 - (b) mistakes in DNA replication.
 - (c) translocations and mistakes in meiosis.
 - (d) recombination at fertilization.
4. Charles Darwin was the first person to propose
 - (a) that humans evolved from apes.
 - (b) that evolution occurs.
 - (c) that evolution occurs through a process of gradual change.
 - (d) that the Earth is older than a few thousand years.
 - (e) a mechanism for how evolution occurs.
5. How can you determine whether or not a given population is in genetic equilibrium?
 - (a) Compare gene frequencies and genotype frequencies of two generations.
 - (b) Evaluate the population's response to environmental stress
 - (c) Assess the mutation rates of the population's genes
 - (d) Compare the population size over generations
 - (e) Calculate the ratio of males to females in the population

True / False (10 marks)

Answer True or False

6. True or False: If messenger RNA molecules were not destroyed, they would replicate rapidly, leading to excessive polypeptide production.
7. True or False: Autoradiography can effectively demonstrate the incorporation of amino acids into proteins within cells.
8. True or False: Normal somatic cells of a human male do not contain an inactivated X chromosome due to the presence of only one X chromosome.
9. True or False: Viruses are not independent organisms and require a host cell to replicate.
10. True or False: The symplastic pathway for sucrose movement includes fibers, bundle sheath, companion cell, and phloem parenchyma.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the emergence of the first mammals during the Carboniferous Period and analyze the key differences between monotremes and marsupials in terms of their reproductive strategies and adaptations.
12. Discuss the challenges associated with classifying the tissues of higher plants, highlighting the role of evolutionary heritage in their characterization.

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